

Fernando Moreira da Silva

Full-Stack TypeScript (React/Next.js) · Application Security (OWASP) · Author of Nemesis Defender

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Portfolio: hub-fernando-dev.vercel.app · GitHub: github.com/feryamaha · LinkedIn: linkedin.com/in/feryamaha

PROFESSIONAL SUMMARY

Full-stack TypeScript developer (React/Next.js) focused on clean architecture, security by default, and AI-assisted development under deterministic governance (a custom multi-IDE harness). Author and maintainer of Nemesis Defender, an open-source security enforcement framework written in Rust with a kernel layer in eBPF/LSM on Linux. I apply OWASP, runtime validation, and supply-chain security in day-to-day production. I moved into software in 2022, after 17 years in industrial metrology, from which I bring explicit contracts, low tolerance for ambiguity, and verifiable quality. Currently pursuing a Software Engineering degree.

TECHNICAL SKILLS

Languages: TypeScript (strict mode), JavaScript (ES6+), Rust, SQL, HTML5, CSS3

AI-Assisted Development & Agentic Coding: Claude Code, Codex, Cursor, Windsurf, Devin, Antigravity, OpenCode, Grok, Gemini; LLM-agent governance via a custom multi-IDE harness: SDD pipeline (spec, critical analysis, plan, implementation, tests and doc-sync), deterministic pretool hooks (exit 2), subagent-driven development, trust ledger and versioned canonical rules

Frameworks & Libraries: React, Next.js (App Router, React Server Components), Node.js (BFF via Route Handlers), React Hook Form, Zod, Tailwind CSS, Recharts, next-intl (i18n), Framer Motion

Architecture: Clean Architecture, Design Systems, BFF (Backend for Frontend), layered separation (UI, Hooks, Services, Types), explicit TypeScript contracts

APIs & Data: REST API integration and aggregation with provider fallback, caching, rate limiting

Databases: PostgreSQL, Drizzle ORM (schema and migrations via drizzle-kit), Neon (serverless Postgres), data modeling

Security: OWASP, CSP Level 3 (dynamic nonce), HSTS, X-Frame-Options, Permissions-Policy, authentication and session (Auth.js/NextAuth, bcrypt hashing), hardening, threat modeling, runtime validation

DevSecOps & Supply-Chain: working knowledge of SLSA (provenance attestation), CI audit gate (cargo audit), SHA-256 checksums, signed commits (DCO)

Systems & Low-Level: eBPF, BPF LSM (Linux kernel), POSIX contract (exit codes, stdio), Linux/macOS terminal, AST analysis (tree-sitter), static analysis, fail-closed design

Performance: RSC, SSR, SSG, ISR, bundle optimization, LCP/TTI metrics, Lighthouse

Testing & Quality: Playwright (E2E), ESLint and Biome (custom rules), strict typing, Zod validation, regression tests

DevOps & Tools: Git, GitHub Actions (CI), Cargo, Bun, npm, Yarn, Linux, Figma

PROJECTS

Nemesis Defender (release v0)

Author and maintainer · Rust + eBPF · AGPL-3.0 · github.com/feryamaha/Nemesis_Defender_v0

- Independently designed and implemented a three-layer security enforcement framework (pre-tool hooks, content scanner, and eBPF LSM in the Linux kernel) that blocks destructive commands and supply-chain malware before execution.
- Built over roughly 1.5 years inside real production projects and open-sourced after validating the thesis; during that period, no AI agent executed a destructive command or accessed a sensitive file without permission.
- Implemented a 6-stage scanning pipeline (AST via tree-sitter, byte-level, regex, deny-list, entropy, and recursive decoder), combining a curated deny-list and semantic-analysis visitors against prompt injection, decode-then-exec, credential exfiltration, reverse shells, and typosquatting.
- Developed the kernel layer in eBPF/BPF LSM (bprm_check_security and egress control via socket_connect) as an independent containment net for destructive commands, with local logging to a violations ledger.
- Validated the system with adversarial red-teaming using real LLM agents and a custom suite of 184 tests across 26 modules, including a module dedicated to measuring false positives; every bypass found was fixed and covered by regression tests.
- Structured the release pipeline with SHA-256 checksums, provenance attestation (SLSA), and a CI audit gate (pentest and cargo audit) before each release, with blocking rules embedded in the binary (tamper-proof).
- Applied fail-closed design, corroboration-model quarantine, and IDE-agnostic architecture (Claude Code, Codex, Cursor, VS Code, Devin), validated on Linux and macOS, with complete technical documentation under AGPL-3.0.

PROFESSIONAL EXPERIENCE

Full-Stack Developer (TypeScript)

Nov 2024 - Present

- Developed web applications, dashboards, and portals with Next.js (App Router), React, and TypeScript
- Implemented a BFF via Route Handlers, keeping credentials and sensitive rules exclusively on the server.
- Structured the frontend architecture in layers (UI, Hooks, Services, Types) with explicit contracts and strict separation of responsibilities.
- Applied security by default aligned with OWASP: CSP Level 3 with dynamic nonce, HSTS, X-Frame-Options, and runtime validation with Zod.
- Built a Design System in Tailwind CSS with semantic tokens and TypeScript-typed components, accelerating feature delivery.
- Optimized performance (RSC, SSR, SSG, ISR) with analytics monitoring and Lighthouse score improvements
- Led AI-assisted development (Devin/Windsurf, Claude, Codex, Cursor) under explicit architectural governance: features, refactors, fixes, and documentation.

Laboratory Technician / Metrologist2007 - 2024

Unipac / Síntegra Surgical Sciences · Pompeia, SP

- 17.5 years in dimensional metrology with advanced GD&T, technical-drawing interpretation, and reverse engineering, ensuring millimeter precision and quality compliance.
- Developed measurement programs for CMM (Mitutoyo, PolyWorks, PC-DMIS) and MSA and SPC studies, APQP Core Tools.
- A foundation of technical discipline that shapes how I build software: explicit contracts, low tolerance for ambiguity, and verifiable quality.

EDUCATION

Bachelor's in Software Engineering (in progress)Jan 2026

Universidade Estácio de Sá

Self-Taught Program in Software EngineeringNov 2022 - Present

Frontend Software Engineering — Architecture, Security, and AI Governance

Self-directed study, without formal instruction: I analyzed the curricula of Software Engineering programs and studied on my own, organizing readings, research, study time, and practice. Mastery built through intensive hands-on experience and solving real problems in digital products, focused on frontend architecture, advanced patterns, critical analysis of complex systems, internal frameworks, design systems, applied security, and AI governance.

Technical Diploma in Plastics2016 - 2018

SENAI Shunji Nishimura

CERTIFICATIONS & CONTINUING EDUCATION

- AI Governance Professional · Securiti.ai
- OWASP Top 10 for LLM Applications · OWASP Foundation
- AWS Cloud Fundamentals · AWS Skill Builder
- Self-taught in Software Engineering since 2022 (performance, design systems, and security architecture).

LANGUAGES & AVAILABILITY

Languages: Portuguese (native) · English (technical reading; improving) · Spanish (intermediate)

Availability: Full-time (CLT), contractor (PJ), or freelance · Remote or hybrid